

Sample Superstore Sales Analysis Dashboard

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1. Project Overview and Objective

This project involves cleaning, transforming, and analyzing raw data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful Sales Analysis business insights.

2. Data Sources.

- **Source Description and Timeline:** Data government/ World Bank Open Data/ UCI/ Google Dataset Search 2020-2025.
- **Domain:** Sample Superstore Sales Analytics etc.

3. Problem Statement.

- To analyse sales performance across regions and products to identify top-performing areas and profit trends.
- To study crop production patterns across states and years for better agricultural planning.
- To understand customer purchasing behavior to enhance marketing and retention strategies.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Product Name	Text (String/ Categorical)	Identify and Analyze product
Order Date	Date/Time	Used For Time based Analysis
Shipment Date	Date/ Time	The Date on which the order was shipped
Shipment Mode	Text / Categorical	The shipping method used to deliver the order
Country Region	Text / Categorical	The countru where the order was placed and delivered

Sample Superstore Sales Analysis Dashboard

State	Text / Categorical	The state of where the customer is located
Category	Text / Categorical	The main clasification of products
Sub-Category	Text / Categorical	The more detailed clasification within a category
Sales	Numeric / Decimal Number Currency	The total revenue generated from the sale of product
Quantity	Numeric / Whole Number Integer	The number of unit sold in a particular order
Discount	Numeric / Decimal Number	The reduction applied to the products selling price
Profit	Numeric / Decimal Number Currency	The earnings generated after deducting costs from sales

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and Analysis sales initial insights.

7. Data Modelling and DAX (Power BI)

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- **Data Model: My Project** used single-table dataset, so relationships were not required
- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, total profit, total Quantity, sum of discount and Profit margin.created columns by shipping days, and years.

Shipping Days = DATEDIFF([Order-Date], [Ship-Date], DAY)

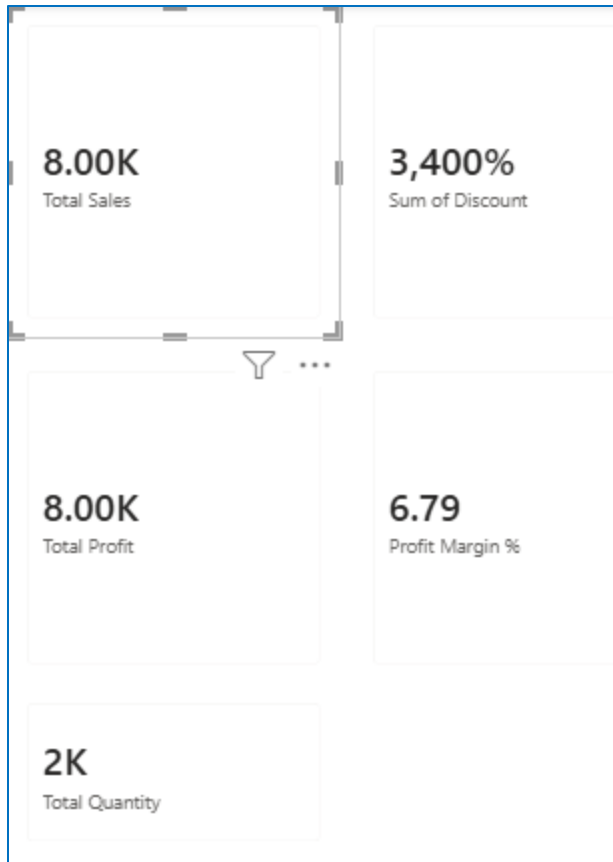
Shipping Days
122
122
30
122
122
122
61
122
122
-259
-259
-291
3

Year = YEAR([Order-Date])

Year
2023
2023
2023
2023
2023
2023
2023
2023
2023
2023
2023

Measures :

Sample Superstore Sales Analysis Dashboard

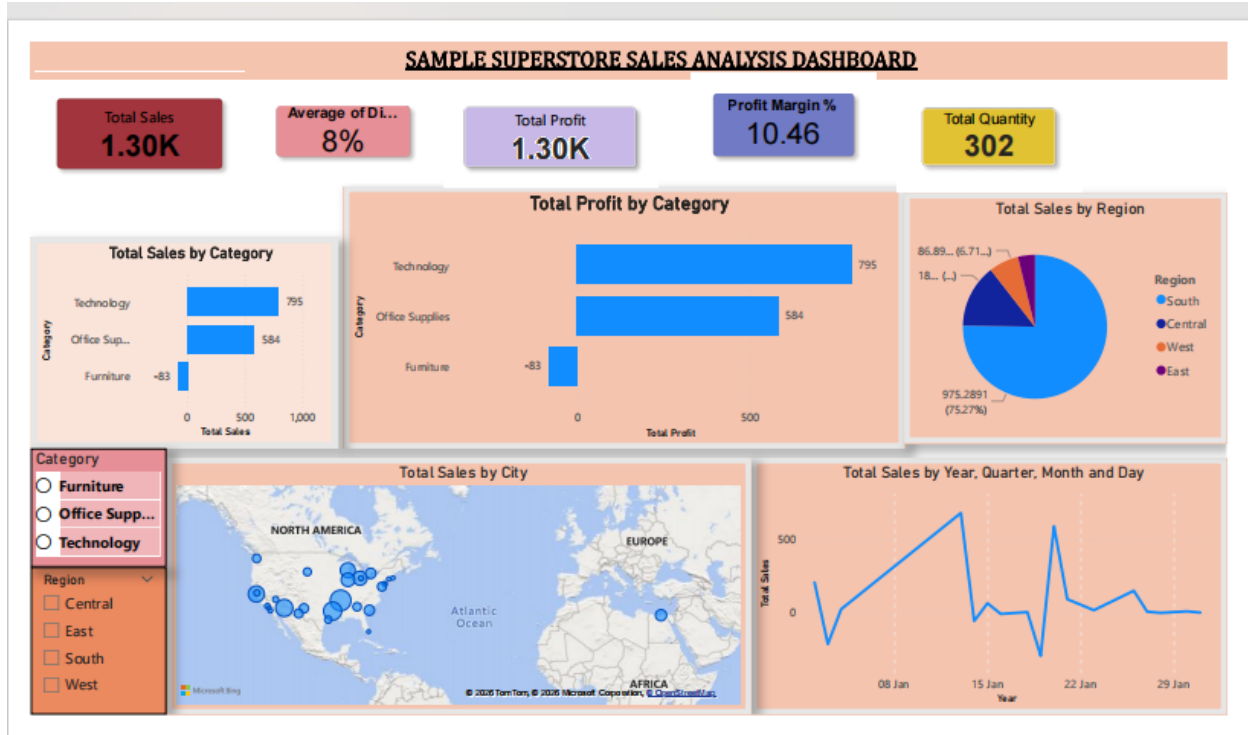


8. Analysis and Visualizations (Power BI)

Dashboard Features:

Multiple Visualization: Created pie chart, donut chart, map chart, bar chart, column chart and slicers.

Sample Superstore Sales Analysis Dashboard



9. Insights & Conclusions

Overall business performance:

The business is maintaining a healthy profit margin of nearby 20%, indicating good profitability from sales.

Category Performance:

Technology is the top-performing category, contributing about 45% of total sales.

Furniture generates the lowest sales and may require promotional strategies or product optimization.

Focusing on Technology products can further boost revenue.

Regional Analysis:

The west region is the strongest market, generating over one-third of total sales.

The south region has the lowest contribution and presents an opportunity for growth initiatives.

Monthly sales trends:

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Sales increased significantly from january to march.

A dip occurred in may

Strong recovery was observed in july and october

October recorded the highest sales (400)

Sales declined during november and december.

10. Conclusions.

The Superstore Sales Analysis Dashboard was developed using Power Bi to analysis sales performance, profitability, Product categories, regional contributions, and sales trends. The dashboard provides an interactive and userfriendly view of key business metrics, enabling better decision-making through data visualization.